CookAnything: Flexible and Consistent Single-Pass Recipe Image Generation

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Abstract

Generating step-by-step recipe illustrations requires more than prompt-following in isolated images: a model must map each instruction to the correct visual state, preserve ingredient identity across steps, and remain stable when the number of steps varies from recipe to recipe. Existing text-to-image systems are not designed for this setting, and naively concatenating all steps often causes semantic entanglement, repeated content, and inconsistent ingredients. We present CookAnything, a diffusion-transformer framework built on FLUX.1-dev that generates an arbitrary-length sequence of recipe images in a single denoising pass. CookAnything combines Step-wise Regional Control, which binds each step to a dedicated latent region while mixing in a global recipe prompt, Flexible RoPE, which resets positional indices across step regions to avoid long-range spatial drift, and Cross-Step Consistency Control, which fuses per-step text with contextual tokens extracted from the full recipe and Cooking-Agent-enriched ingredient descriptions. On RecipeGen, the training-based variant achieves the best Goal Faithfulness (30.59), CLIP-based Step Faithfulness (30.45), GPT-based Step Faithfulness (8.69), Ingredient Accuracy (9.27 GPT, 7.15 human), and human Usability (8.48), while the training-free variant yields the best Cross-Step Consistency (0.17) and GPT-based Usability (9.89). On VGSI-Recipe, CookAnything also generalizes strongly, with the training-free variant reaching 31.22 Goal Faithfulness and 1.67 Cross-Step Consistency. Human evaluation further shows a large perceptual advantage over FLUX.1-dev and SD3.5, supporting CookAnything as an effective framework for coherent procedural food illustration.

CCS Concepts

• **Computing methodologies** → **Image generation**; *Neural networks*.

Keywords

recipe image generation, diffusion transformers, regional attention, procedural image generation, food computing

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Figure 1: CookAnything generates all recipe steps in a single pass while supporting different recipe lengths and dishes. The output is a vertically stacked multi-step canvas that is later split into individual instructional images.

1 Introduction

Recipes are naturally procedural and multimodal: users reason about a target dish, a sequence of state-changing steps, and the ingredient transformations that connect them. Large-scale food datasets and food-computing systems have improved recipe understanding, retrieval, and generation [7, 10, 13, 15], but step-wise recipe image generation remains relatively underexplored. Recent work on illustrated instructions and recipe image generation highlights the need for visual faithfulness, cross-step coherence, and user-facing instructional quality [14, 28]. Those requirements are stricter than standard text-to-image generation because a successful model must depict not only plausible food images, but the *right intermediate state* for each step.

This setting exposes a mismatch between recipe generation and standard diffusion pipelines. If each step is generated independently, ingredient identity and visual style drift across steps. If all steps are concatenated into one long prompt, different sub-instructions compete for attention, often leading to ingredient leakage, duplicated content, or misplaced step content. The problem becomes harder when the number of steps is not fixed: many consistency-oriented multi-image methods assume a constant layout or perform iterative generation, which is inconvenient for recipe workflows that range from a few steps to long procedures [8, 14, 30].

We address this problem with *CookAnything*, a diffusion-transformer framework for arbitrary-length recipe illustration in a single denoising pass. The core idea is to align each textual step to its own latent region, while still injecting recipe-level context so that ingredients and global appearance remain coherent. Concretely, we generate one tall latent canvas whose height scales with the number of steps. Step-wise Regional Control (SRC) restricts each step prompt to the

corresponding latent stripe through an attention mask, and interpolates regional features with a whole-recipe base prompt to preserve global coherence. Flexible RoPE resets positional indices across step regions, preventing the positional drift that arises when standard rotary embeddings treat a long stacked canvas as one monolithic image. Cross-Step Consistency Control (CSCC) further stabilizes ingredient continuity by enriching step prompts with contextual tokens from the full recipe and by using a Cooking Agent to make implicit ingredients explicit and phrased consistently across steps.

CookAnything is motivated by a simple hypothesis: recipe image generation improves when local step grounding and global recipe context are enforced simultaneously, rather than traded off against each other. The provided implementation supports both a training-free setting, where the method is applied directly on FLUX.1-dev, and a training-based setting, where the same framework is paired with LoRA adaptation. Across both settings, CookAnything outperforms strong baselines on RecipeGen and VGSI-Recipe, and human raters strongly prefer its outputs.

Our contributions are threefold:

- We present a unified DiT-based framework for arbitrary-length, joint recipe image generation in a single denoising pass.
- We introduce Step-wise Regional Control, Flexible RoPE, and Cross-Step Consistency Control to disentangle step semantics while preserving cross-step ingredient continuity.
- We show strong results in both training-free and training-based regimes on RecipeGen and VGSI-Recipe, including the best overall RecipeGen performance and a large human-evaluation margin over strong DiT baselines.

2 Related Work

2.1 Food and Recipe Generation

Food computing spans recognition, retrieval, nutrition analysis, and recipe understanding [4, 5, 13, 15]. Several recent systems broaden this space with foundation-model or multimodal-generation capabilities, including FoodLMM [27], ChefFusion [10], and FIRE [7]. These works improve recipe-text generation or general food understanding, but they do not directly solve the problem studied here: generating a coherent *sequence* of step images from a recipe. Datasets such as MM-ReS and MIAIS emphasize that recipes are inherently sequential and ingredient-aware [16, 29], while RecipeGen makes step-wise recipe image generation a direct benchmark [28]. CookAnything targets this procedural visual-generation setting explicitly.

2.2 Illustrated Instructions and Multi-Image Consistency

Generating illustrated instructions has recently emerged as a distinct task. StackedDiffusion [14] formalizes evaluation criteria such as validity and consistency, but its pipeline is not built around arbitrary-length, single-pass recipe generation. More broadly, story and multi-turn generation methods such as StoryDiffusion [30], StoryDALL-E [12], and AutoStudio [6] focus on long-range subject consistency in narratives or interactive editing. They are informative for cross-image coherence, yet recipe illustration differs in two important ways: step order is semantically constrained by state transitions, and the salient consistency target is often *ingredients* rather than characters. CookAnything specializes consistency control for this setting by coupling step-local grounding with recipe-global context fusion.

2.3 Regional and Layout Control in Diffusion Models

Regional control has been explored for improving compositional faithfulness in diffusion models. GLIGEN [11], BoxDiff [25], and attention-refocusing methods [18] improve layout controllability, while regional prompting for DiTs extends such ideas to FLUX-like backbones [3]. These methods usually target object placement within a single image. CookAnything instead uses region control to align *procedural steps* with fixed latent stripes in a joint canvas, then mixes the resulting regional features with a global base prompt to retain recipe-level coherence. This repurposes regional prompting from spatial composition to sequential procedural generation.

3 Method

3.1 Problem Setup

Given a recipe with goal description g and ordered step texts $\{s_i\}_{i=1}^{N}$, our goal is to generate a sequence of images $\{x_i\}_{i=1}^{N}$, where each image depicts the visual state corresponding to one step and the full sequence remains coherent. Unlike pipelines that generate each x_i separately, CookAnything produces the sequence jointly on a single latent canvas. For a base resolution $W \times W$, we allocate a canvas of size $W \times (NW)$ and split it into N horizontal stripes, one per step. After denoising, the decoded canvas is split back into step images.

CookAnything is built on FLUX.1-dev [9], a diffusion transformer derived from the DiT paradigm [17]. The model operates in latent space as in modern latent diffusion systems [19, 22, 24]. Our modifications are inference-time compatible and are also used in a training-based LoRA variant.

3.2 Step-wise Regional Control

The first challenge is semantic disentanglement: different step descriptions should affect only the visual region corresponding to that step. Let M_i denote the binary mask for stripe i. CookAnything builds a regional attention mask that permits text tokens from step i to attend to latent tokens inside M_i, and latent tokens in M_i to attend back to the same step tokens. Latent tokens may also attend within the same region and to uncovered background positions. This is implemented by constructing a block attention mask over the concatenated text and image tokens.

We do not rely on purely regional conditioning. A second prompt branch encodes a *base prompt* describing the full recipe goal (for example, the dish name). During the early denoising stages, CookAnything computes both regional and global attention outputs and linearly interpolates them:

$$\mathbf{h}^{\text{mix}} = (1 - \beta)\mathbf{h}^{\text{reg}} + \beta\mathbf{h}^{\text{base}}, \tag{1}$$

where β is the base-ratio hyperparameter. In the provided implementation, this global interpolation is applied only in the masked

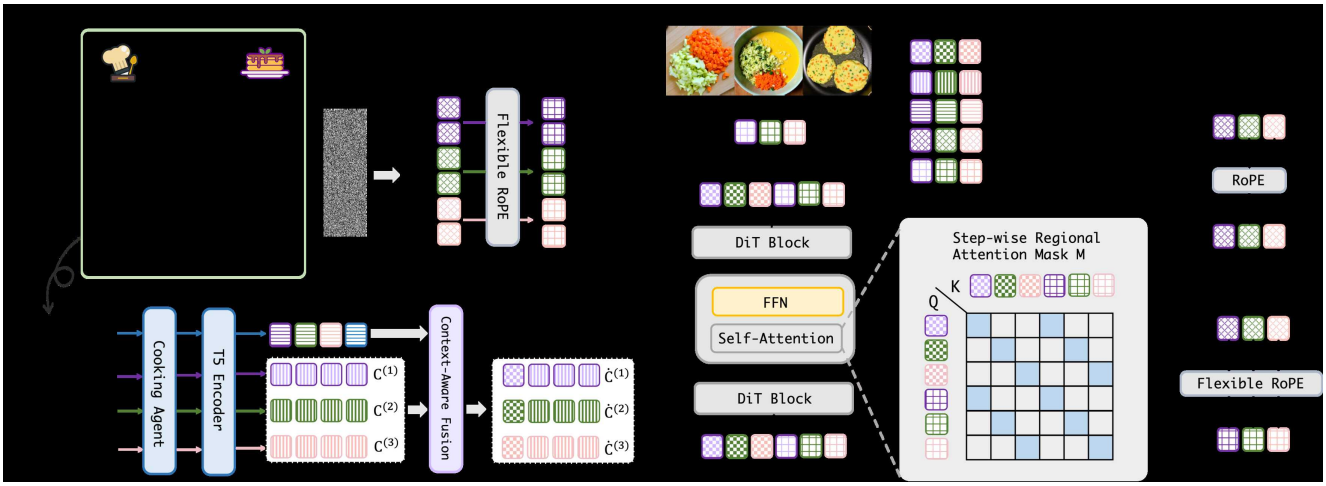

Figure 2: Overview of CookAnything. A recipe is mapped to a vertically stacked latent canvas, where each step occupies a dedicated region. Step-wise Regional Control applies regional attention masks and mixes regional outputs with a global base prompt; Flexible RoPE resets positional indices across step regions; and Cross-Step Consistency Control enriches step embeddings with recipe-level contextual tokens and Cooking-Agent guidance.

early stages, then the model falls back to the base prompt alone later in denoising. This design reflects an intuitive division of labor: early steps establish which content belongs to which region, while later steps refine the shared appearance of the full recipe canvas.

3.3 Flexible RoPE

A tall stacked canvas introduces an unusual positional-encoding problem. Standard rotary position embeddings treat the entire canvas as one continuous image, so the positional indices for lower steps drift far away from those at the top, even though each step semantically occupies an analogous local layout. The result is positional entanglement: lower-step regions may inherit distorted or unstable spatial behavior.

CookAnything addresses this with Flexible RoPE. Instead of assigning strictly monotonic image indices along the full vertical axis, we reset the positional coordinates within each step stripe. Conceptually, for a pixel or latent location (x, y) on the stacked canvas, we use

$$\pi(x, y) = (x,\ y \bmod W), \tag{2}$$

so that every step region reuses the same local coordinate frame. In implementation terms, latent image identifiers are prepared for a single square region and then repeated across the number of step regions. This preserves long-range stability while keeping each step spatially aligned to a shared local template.

3.4 Cross-Step Consistency Control

Semantic disentanglement alone does not guarantee ingredient continuity. Recipes often omit ingredients in later steps once they have been introduced, and the same ingredient may be referred to differently over time. To reduce such ambiguity, CookAnything uses a Cooking Agent to rewrite the recipe into explicit, image-oriented step descriptions. The agent is instructed to enumerate ingredients, preserve shapes and cuts, avoid future ingredients, and maintain consistent wording across steps.

After the rewritten recipe is obtained, we encode both each local step prompt and the full concatenated recipe with a T5-family text encoder [21]. For each step i, we extract a contextual token span $\mathbf{c}_i$ from the full-recipe encoding and fuse it with the local step embedding $\mathbf{e}_i$:

$$\tilde{\mathbf{e}}_i = \mathbf{e}_i + \lambda \mathbf{c}_i, \tag{3}$$

where λ is a fixed fusion weight. In the released implementation, $\lambda = 0.2$. This additive fusion allows a step prompt to retain its local semantics while inheriting information about ingredient identity and naming from the full recipe. We refer to this mechanism as Cross-Step Consistency Control (CSCC).

3.5 Training-Free and Training-Based Variants

CookAnything supports two deployment modes. In the training-free (TF) variant, all controls are applied directly to pretrained FLUX.1-dev. In the training-based (TB) variant, the same generation framework is combined with LoRA adaptation, as reflected by the inference pipeline's optional LoRA loading. Both variants share the same joint-generation mechanism and differ only in whether recipe-specific adaptation is used.

4 Experiments

4.1 Datasets, Baselines, and Metrics

We report results on RecipeGen [28], a benchmark designed for real-world recipe image generation, and VGSI-Recipe, a recipe-oriented adaptation of the Visual Goal-Step Inference setting [26]. RecipeGen evaluates goal, step, ingredient, and usability quality on real recipes, while VGSI-Recipe tests generalization to a distinct procedural benchmark.

Baselines cover three families: strong single-image generators such as SD1.5/2.1, SDXL, SD3.5, and FLUX.1-dev [9, 19, 22–24]; joint

or in-context generation methods such as SKD, StoryDiffusion, IC-LoRA, and RPF [3, 8, 14, 30]; and layout-aware methods such as GLIGEN and attention refocusing [11, 18]. We evaluate Step Flexibility, Joint Generation, Goal Faithfulness (GF), Step Faithfulness based on CLIP and GPT-style evaluation (SF(C), SF(G)), Cross-Step Consistency (CSC), Ingredient Accuracy (IA), and Usability (UB). Higher is better except for CSC. Table 1 summarizes the metrics.

Table 1: Evaluation metrics for CookAnything.

Metric	Description
Step Flexibility	Indicates whether the model can generate a variable number of step images in a single pass.
Joint Generation	Indicates whether the model can generate all step images simultaneously to ensure consistency and efficiency.
Goal Faithfulness	CLIP [20] similarity between the final image and last-step caption, measuring alignment with the overall goal.
Step Faithfulness	Assesses each image's alignment with its step caption using CLIP and a GPT-based evaluator [1].
Cross-Step Consistency	Based on StackedDiffusion [14] and DINO [2], uses l_2 distance and step count difference to assess visual and numerical consistency.
Ingredient Accuracy	Uses a GPT-based evaluator and manual inspection to verify whether the expected ingredients are visually present in each step and to check for any omissions.
Usability	Evaluates spatial alignment in jointly generated images to avoid layout collapse or misalignment, using a GPT-based evaluator and human inspection. The evaluator scores usability across five aspects: size consistency, step clarity, content duplication, process completeness, and step count reasonableness.

4.2 Implementation Details

The released code generates a single stacked image whose width is fixed to 512 and whose height is computed as $512 \times N$ for an N-step recipe. Step masks are contiguous 512×512 stripes extracted from the ordered recipe text. Inference uses 24 denoising steps with guidance scale 3.5, applies regional masking for the full denoising trajectory in the default script, and mixes regional and global features with base ratio 0.1. The contextual fusion weight in CSCC is fixed to $\lambda = 0.2$, and the default implementation injects the regional controls at every transformer block. The training-based variant loads a LoRA adapter on top of FLUX.1-dev, while the training-free variant keeps the pretrained model frozen. We report the results provided with the project artifacts and do not infer additional hyperparameter sweeps beyond those released.

4.3 Main Results on RecipeGen

Table 2 reports the main comparison on RecipeGen. CookAnything is the only method among the strongest performers that simultaneously supports variable step counts and joint generation. The training-based variant achieves the best GF (30.59), SF(C) (30.45), SF(G) (8.69), IA(G/H) (9.27/7.15), and UB(H) (8.48). The training-free variant attains the best CSC (0.17) and UB(G) (9.89), while remaining competitive on all other metrics.

Two comparisons are particularly informative. First, against strong single-image DiT backbones such as SD3.5 and FLUX.1-dev, CookAnything gains more than 3 points in GF and about 2 points in SF(C), while drastically improving ingredient accuracy and usability. This indicates that the main benefit is not merely a stronger backbone, but the joint procedural generation mechanism. Second, against joint-generation baselines such as SKD, IC-LoRA, and RPF, CookAnything improves both faithfulness and consistency, suggesting that simple image concatenation or in-context tuning is insufficient for recipes without explicit step-region binding. Measured against FLUX.1-dev, the training-based variant improves GF by 4.12, SF(C) by 2.14, and GPT-based IA by 3.34 on RecipeGen; against the joint baseline SKD, it still gains 3.97 GF and 1.92 SF(C). These margins are important because they show that CookAnything is not only better than per-step generation, but also better than joint generation without explicit procedural structure.

4.4 Generalization to VGSI-Recipe

CookAnything also transfers well to VGSI-Recipe, as shown in Table 3. The training-free variant obtains the best GF (31.22), SF(G) (7.12), CSC (1.67), IA (8.42), and UB (9.06), while the training-based variant yields the best SF(C) (29.71). This pattern is consistent with the RecipeGen results: LoRA adaptation slightly improves prompt-image alignment, whereas the pure training-free control stack remains especially strong on cross-step consistency. Importantly, both variants outperform FLUX.1-dev by large margins across all reported axes, indicating that the improvements are not dataset-specific.

4.5 Human Evaluation

Human evaluation in Table 4 shows a large perceptual advantage for CookAnything over SD3.5 and FLUX.1-dev. CookAnything reaches 70.82 GF, 68.18 SF(C), 69.69 CSC, 60.38 AQ, and 69.68 OA, while the best baseline remains below 26.34 on AQ and below 17.61 on the other criteria. The margin is large enough that it is unlikely to be explained by small stylistic preferences alone; raters appear to prefer CookAnything on both procedural clarity and overall image quality.

5 Analysis

5.1 Ablation Study

Table 5 isolates the contribution of each component on RecipeGen. Removing Flexible RoPE causes the largest drop in faithfulness, reducing GF from 30.59 to 28.66 and SF(C) from 30.45 to 29.33. This supports the claim that positional reset is essential when the canvas height grows with the number of steps. Removing CSCC hurts consistency most directly, increasing CSC from 0.19 to 0.29. Finally, removing the Cooking Agent leads to the sharpest collapse in ingredient continuity, driving CSC to 3.06. Interestingly, the variant without CSCC still remains competitive on GF and SF, which suggests that local prompt grounding alone can preserve per-step

Table 2: Comparison with Other Models in RecipeGen. SF means Step Faithfulness, IA means Ingredient Accuracy and UB means Usability. (C) means CLIP score, (G) means GPT-based score and (H) is Human evaluation. The UB metric is applicable only to methods capable of Joint Generation. TF means Training-Free and TB means Training-Based.

Category	Method	Step Flexibility	Joint Generation	Goal Faithfulness ↑	Cross-Step Consistency↓	SF↑		IA↑		UB ↑	
						(C)	(G)	(G)	(H)	(G)	(H)
UNet-based	StoryDiffusion [30]	✓	✗	17.51	6.01	25.54	5.30	7.25	3.41	—	—
	SDXL [19]	✓	✗	27.46	2.98	29.37	6.79	7.51	3.71	—	—
	SKD [14]	✗	✓	26.62	0.7	28.53	4.57	6.67	2.59	6.43	3.59
DiT-based	SD3.5 [24]	✓	✗	27.42	2.97	28.77	6.73	7.58	3.97	—	—
	Flux.1-dev [9]	✓	✗	26.47	3.47	28.31	5.31	5.93	5.71	—	—
	IC-LoRA [8]	✓	✓	26.07	9.03	26.58	4.03	5.50	3.91	5.34	4.45
	RPF [3]	✓	✓	27.19	8.73	25.99	4.45	7.05	3.02	3.89	4.24
Layout-aware	GLIGEN [11]	✓	✗	26.99	2.17	26.72	5.17	6.16	5.28	—	—
	A-R [18]	✓	✗	26.31	2.46	27.63	4.58	5.46	5.29	—	—
DiT-based	Ours (TF)	✓	✓	30.12	**0.17**	29.80	8.52	9.12	6.92	**9.89**	7.66
	Ours (TB)	✓	✓	**30.59**	0.19	**30.45**	**8.69**	**9.27**	**7.15**	9.70	**8.48**

Table 3: Comparison with Other Models in VGSI-Recipe. SF (C) and SF (G) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF↑	SF(C)↑	SF(G)↑	CSC↓	IA↑	UB↑
SD1.5 [22]	27.03	26.63	3.41	12.28	5.32	—
SD2.1 [23]	27.03	26.63	3.32	11.94	5.01	—
SDXL [19]	27.78	28.32	3.60	11.7	5.87	—
SD3.5 [24]	26.12	27.03	3.43	5.27	5.17	—
Flux.1-dev [9]	25.82	27.77	2.99	5.97	4.55	—
IC-LoRA [8]	26.18	28.25	3.77	4.88	5.74	6.06
SKD [14]	28.25	28.26	4.22	3.14	6.35	5.75
RPF [3]	28.40	26.54	3.25	7.12	6.03	3.96
GLIGEN [11]	29.70	29.48	5.01	4.08	6.93	—
A-R [18]	28.87	28.34	4.32	3.94	6.31	—
Ours (TF)	**31.22**	29.61	**7.12**	**1.67**	**8.42**	**9.06**
Ours (TB)	29.88	**29.71**	6.63	2.26	8.07	7.72

Table 4: Human Evaluation on Perceptual Quality.

Method	GF ↑	SF(C)↑	CSC ↑	AQ ↑	OA↑
SD3.5	12.08	15.60	12.70	13.58	13.46
Flux.1-dev	17.11	16.23	17.61	26.34	16.86
Ours	**70.82**	**68.18**	**69.69**	**60.38**	**69.68**

semantics reasonably well, but not the cross-step ingredient identity that recipes require.

5.2 Qualitative Analysis

Figure 3 compares CookAnything with representative baselines. The failure modes are consistent with the quantitative results: FLUX.1-dev and SD3.5 often produce visually plausible individual images, but they drift across steps or insert incorrect ingredients;

Table 5: Ablation Study. F-RoPE refers to the Flexible RoPE we proposed, CSCC stands for Cross-Step Consistency Control, and C-Agent refers to the Cooking Agent. SF (C) and SF (G) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF ↑	SF(C)↑	SF(G)↑	CSC ↓
w/o F-RoPE	28.66	29.33	7.93	0.23
w/o CSCC	30.57	30.28	8.67	0.29
w/o C-Agent	29.00	29.59	7.84	3.06
Ours	**30.59**	**30.45**	**8.69**	**0.19**

SKD maintains some global structure yet misses step-specific ingredients; CookAnything, especially the training-based variant, preserves ingredient identity and step progression more faithfully.

The training-free versus training-based comparison also reveals a useful trade-off. On both RecipeGen and VGSI-Recipe, the training-based variant is stronger on prompt-image alignment metrics such as GF and SF(C), while the training-free variant is often best on CSC and sometimes on usability. This suggests that recipe-specific adaptation improves faithfulness to the target step text, but the frozen-backbone variant retains particularly stable cross-step behavior when paired with our inference-time controls. In practice, this makes CookAnything useful in both data-rich and data-limited regimes.

Figure 4 focuses on CSCC. Before contextual fusion, ingredients that should persist across steps may disappear or change form. After CSCC, the same ingredients are depicted more consistently, especially when later-step prompts omit details that were explicit only earlier in the recipe. This behavior matches the ablation trend in CSC.

Figure 5 illustrates the motivation for Flexible RoPE. Under the original positional encoding, lower regions in the stacked canvas can inherit unstable layouts and cross-step confusion. Resetting

Figure 3: Qualitative comparison on RecipeGen examples. Baselines often exhibit missing ingredients, wrong ingredients, or unstable cross-step progression. CookAnything preserves ingredient identity and procedural progression more reliably, with the training-based variant producing the strongest overall fidelity.

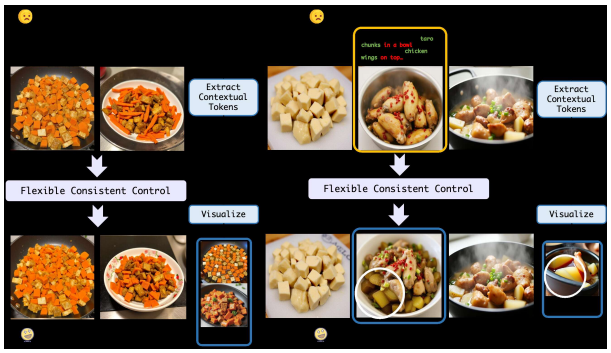

Figure 4: Effect of Cross-Step Consistency Control. Contextual token fusion improves ingredient continuity and reduces omissions or ingredient substitutions across the sequence.

the positional frame per step yields better local alignment and reduces entanglement between distant steps, which explains the large degradation observed when Flexible RoPE is removed.

6 Conclusion

We presented CookAnything, a diffusion-transformer framework for flexible and consistent multi-step recipe image generation. The method combines step-wise regional attention, positional reset across step regions, and recipe-level contextual fusion so that all step images can be generated jointly in a single denoising pass. Across RecipeGen and VGSI-Recipe, CookAnything delivers strong results in both training-free and training-based settings, with particularly large gains in step faithfulness, ingredient accuracy, usability, and human preference. The ablations and qualitative examples support the central conclusion of the paper: for procedural food illustration, explicit step-region alignment and cross-step ingredient context are both necessary for coherent generation.

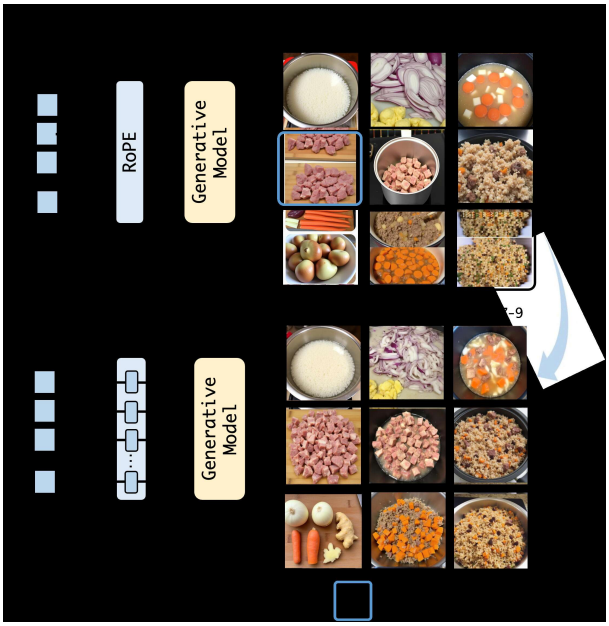

Figure 5: Motivation for Flexible RoPE. Standard rotary embeddings treat the stacked canvas as one long image, which entangles distant steps; resetting the positional frame per step stabilizes local layouts.

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